Student: Xirong Zhang **Date:** 11/30/25	**Instructor:** Dmitriy Ivanov **Course:** Math 120 Fall 2025	**Assignment:** Section 1.2

1. Determine which matrices are in reduced echelon form and which others are only in echelon form.

a. $\begin{bmatrix} 1 & 1 & 0 & 1 & 1 \\ 0 & 2 & 0 & 2 & 2 \\ 0 & 0 & 0 & 5 & 5 \\ 0 & 0 & 0 & 0 & 4 \end{bmatrix}$ **b.** $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$ **c.** $\begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

Is matrix **a** in reduced echelon form, echelon form only, or neither?

- [x] echelon form only
- [ ] reduced echelon form
- [ ] neither

Is matrix **b** in reduced echelon form, echelon form only, or neither?

- [x] reduced echelon form
- [ ] echelon form only
- [ ] neither

Is matrix **c** in reduced echelon form, echelon form only, or neither?

- [ ] reduced echelon form
- [ ] echelon form only
- [x] neither

2. Row reduce the matrix to reduced echelon form. Identify the pivot positions in the final matrix and in the original matrix, and list the pivot columns.

$$\begin{bmatrix} 1 & 2 & 4 & 3 \\ 2 & 4 & 6 & 2 \\ 4 & 6 & 8 & 0 \end{bmatrix}$$

Row reduce the matrix to reduced echelon form and identify the pivot positions in the final matrix. The pivot positions are indicated by bold values. Choose the correct answer below.

A.
$$\begin{bmatrix} \mathbf{1} & 0 & 0 & -1 \\ 0 & \mathbf{1} & 0 & -2 \\ 0 & 0 & \mathbf{1} & 2 \end{bmatrix}$$

B.
$$\begin{bmatrix} \mathbf{1} & 0 & 0 & 0 \\ 0 & \mathbf{1} & 0 & 0 \\ 0 & 0 & \mathbf{1} & \mathbf{1} \end{bmatrix}$$

C.
$$\begin{bmatrix} \mathbf{1} & 0 & 0 & -3 \\ 0 & \mathbf{1} & 0 & \mathbf{1} \\ 0 & 0 & \mathbf{1} & 0 \end{bmatrix}$$

D.
$$\begin{bmatrix} \mathbf{1} & 0 & 0 & -3 \\ 0 & \mathbf{1} & 0 & 2 \\ 0 & 0 & \mathbf{1} & -2 \end{bmatrix}$$

Identify the pivot positions in the original matrix. The pivot positions are indicated by bold values. Choose the correct answer below.

A.
$$\begin{bmatrix} \mathbf{1} & 2 & 4 & 3 \\ 2 & \mathbf{4} & 6 & 2 \\ 4 & 6 & \mathbf{8} & 0 \end{bmatrix}$$

B.
$$\begin{bmatrix} \mathbf{1} & 2 & 4 & 3 \\ \mathbf{2} & 4 & 6 & 2 \\ \mathbf{4} & 6 & 8 & 0 \end{bmatrix}$$

C.
$$\begin{bmatrix} \mathbf{1} & 2 & 4 & 3 \\ 2 & \mathbf{4} & 6 & 2 \\ 4 & 6 & \mathbf{8} & \mathbf{0} \end{bmatrix}$$

D.
$$\begin{bmatrix} 1 & 2 & 4 & \mathbf{3} \\ 2 & 4 & \mathbf{6} & 2 \\ 4 & \mathbf{6} & 8 & 0 \end{bmatrix}$$

List the pivot columns. Select all that apply.

- [ ] **A.** Column 4
- [x] **B.** Column 3
- [x] **C.** Column 1
- [x] **D.** Column 2

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3. Find the general solution of the system whose augmented matrix is given below.

$$\begin{bmatrix} 1 & 2 & 1 & 2 \\ 4 & 8 & -1 & -17 \end{bmatrix}$$

Select the correct choice below and, if necessary, fill in the answer box(es) to complete your choice.

A.
$x_1 =$ ____
$x_2 =$ ____
$x_3 =$ ____

B.
$x_1 =$ ____
x_2 is free
x_3 is free

C.
$x_1 = -3 - 2x_2$
x_2 is free
$x_3 = 5$

D. The system has no solution.

4. Find the general solution of the system whose augmented matrix is given below.

$$\begin{bmatrix} 0 & 1 & -2 & 4 \\ 1 & -2 & 0 & -2 \end{bmatrix}$$

Select the correct choice below and, if necessary, fill in any answer boxes to complete your answer.

- **A.** $x_1 = 6 + 4x_3$, $x_2 = 4 + 2x_3$, x_3 is free
- **B.** $x_1 =$ ____, x_2 is free, x_3 is free
- **C.** $x_1 =$ ____, $x_2 =$ ____, $x_3 =$ ____
- **D.** The system has no solution.

5. Find the general solution of the system whose augmented matrix is given below.

$$\begin{bmatrix} 1 & -8 & 0 & -1 & 0 & -9 \\ 0 & 1 & 0 & 0 & -4 & 3 \\ 0 & 0 & 0 & 1 & 3 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Select the correct choice below and, if necessary, fill in the answer boxes to complete your answer.

- **A.** $x_1 =$ ____, x_2 is free, $x_3 =$ ____, x_4 is free, x_5 is free
- **B.** $x_1 =$ ____, $x_2 =$ ____, x_3 is free, $x_4 =$ ____, $x_5 =$ ____
- **C.** $x_1 = 17 + 29x_5$, $x_2 = 3 + 4x_5$, x_3 is free, $x_4 = 2 - 3x_5$, x_5 is free
- **D.** The system is inconsistent.

6. Find the general solution of the system whose augmented matrix is given below.

$$\begin{bmatrix} 1 & 0 & -8 & 0 & -5 & 3 \\ 0 & 1 & 7 & -1 & 0 & 2 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Select the correct choice below and, if necessary, fill in any answer boxes to complete your answer.

A.

$x_1 =$ $3 + 8x_3$

$x_2 =$ $2 - 7x_3 + x_4$

x_3 is free

x_4 is free

$x_5 =$ 0

B.

$x_1 =$

$x_2 =$

x_3 is free

$x_4 =$

$x_5 =$

C.

$x_1 =$

$x_2 =$

$x_3 =$

$x_4 =$

x_5 is free

D.

The system is inconsistent.

7. Determine whether the statement below is true or false. Justify the answer.

The pivot positions in a matrix depend on whether row interchanges are used in the row reduction process.

Choose the correct answer below.

A. The statement is false. The pivot positions in a matrix are determined completely by the positions of the leading entries in the nonzero rows of any echelon form obtained from the matrix.

B. The statement is true. Every pivot position is determined by the positions of the leading entries of a matrix in reduced echelon form.

C. The statement is false. The pivot positions in a matrix depend on the location of the pivot column.

D. The statement is true. The pivot positions in a matrix are determined completely by the positions of the leading entries of each row which are dependent on row interchanges.

8. Determine whether the statement below is true or false. Justify the answer.

A basic variable in a linear system is a variable that corresponds to a pivot column in the coefficient matrix.

Choose the correct answer below.

A. The statement is false. Not every linear system has basic variables.

B. The statement is true. It is the definition of a basic variable.

C. The statement is false. A variable that corresponds to a pivot column in the coefficient matrix is called a free variable, not a basic variable.

D. The statement is true. If a linear system has both basic and free variables, then each basic variable can be expressed in terms of the free variables.

9. Determine whether the statement below is true or false. Justify the answer.

A general solution of a system is an explicit description of all solutions of the system.

Choose the correct answer below.

- **A.** The statement is false. A general solution is the result of an inconsistent system, which has no particular solution.
- **B.** The statement is false. Each different choice of a free variable produces the same solution of the system.
- **C.** The statement is true. After applying the row reduction algorithm and generating a general solution of a system, the rightmost column displays all of the particular solutions of that system.
- **D.** The statement is true. The row reduction algorithm leads directly to an explicit description of the solution set of a linear

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10. Suppose a 3×8 coefficient matrix for a system has three pivot columns. Is the system consistent? Why or why not?

Choose the correct answer below.

- **A.** There is a pivot position in each row of the coefficient matrix. The augmented matrix will have four columns and will not have a row of the form $\begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix}$, so the system is consistent.
- **B.** There is at least one row of the coefficient matrix that does not have a pivot position. This means the augmented matrix, which will have nine columns, must have a row of the form $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$, so the system is inconsistent.
- **C.** There is a pivot position in each row of the coefficient matrix. The augmented matrix will have nine columns and will not have a row of the form $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$, so the system is consistent.
- **D.** There is at least one row of the coefficient matrix that does not have a pivot position. This means the augmented matrix, which will have nine columns, could have a row of the form $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$, so the system could be inconsistent.